

SI 315: Models of Social Information Theory

Silas Falde

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Logistics

- No AI-generated submissions to assignments
- Required attendance at both lectures and discussions
- Two exams, final is non-cumulative
- iClicker both participation and correctness
- Lowest two grades dropped from each of assignments, quizzes, discussion attendance, lecture attendance

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1 Introduction to Networks

1.1 Strong and Weak Ties

- Strong ties: frequent interaction, trust, emotional closeness
- Weak ties: infrequent interaction, limited intimacy
- Strong ties: overlapping social circles
- Weak ties: bridges between disconnected groups
- Weak ties: novel information and opportunities

1.2 Bridges and Local Bridges

- Bridge: an edge whose removal disconnects the network into separate components
- Local bridge: an edge whose endpoints do not have any direct ties to the same neighboring nodes
- Local bridge: removing it increases the distance between its endpoints, giving it a span of at least 3
- Span of a bridge: the distance between the endpoints after the edge is removed
- Bridges and local bridges: often connect otherwise distant or disconnected regions of the network

1.3 Clustering

- Clustering: connected neighbors also connected to one another
- Local clustering coefficient: connected neighbor pairs divided by possible neighbor pairs
- Formula: $C_i = \frac{2e_i}{k_i(k_i-1)}$
- e_i : edges among node i 's neighbors
- Network clustering coefficient: average local coefficient

1.4 Strong Triadic Closure Property

- Two strong ties sharing a node: likely third edge
- Strong ties around a node: mutually connected neighbors
- Open triad: two strong ties without a connecting edge
- Open triads: pressure toward closure
- Closed triads: higher clustering coefficients

1.5 Homophily

- Homophily: the tendency for people to form ties with similar others
- Similarity may involve age, gender, education, location, interests, or beliefs
- Homophily can produce clustered groups with shared characteristics
- Network ties can reinforce existing similarities through social influence
- Observed similarity may result from selection, influence, or both
- Let p and q be the proportions of two groups, with $p + q = 1$
- Under random mixing, the probability of a cross-group tie is $2pq$
- Cross-group tie probability c under random mixing: expected to be $2pq$
- Homophily where the observed cross-group tie probability is less than expected: $c < 2pq$

1.6 Affiliation Networks

- Affiliation network: a network connecting people to the groups, events, organizations, or activities they participate in
- Two-mode network: contains two types of nodes, such as individuals and affiliations
- One-mode projection: connects two people when they share an affiliation
- Shared affiliations can create indirect social ties and opportunities for interaction
- Affiliations can reveal overlapping communities and patterns of participation

1.7 Social Affiliation Networks

- Social affiliation network: represents relationships between people and the groups, events, or organizations with which they are affiliated
- Membership links connect each person to the affiliations they share
- Shared affiliations can create indirect opportunities for social interaction and information flow
- One-mode projection: connects people who share at least one affiliation
- Projection ties may be weighted by the number of affiliations two people share
- Affiliation networks can reveal overlapping communities, central participants, and patterns of group involvement

1.7.1 Closure in Social Affiliation Networks

- Triadic: nodes with a common neighbor tend to form a direct connection, closing the triad
- Group/focal: nodes connected to the same group or affiliation tend to form direct connections, closing the group-based triad
- Membership: nodes are more likely to join groups that their neighbors are already members of

1.8 Signed Networks

- Signed network: a network in which edges have positive or negative signs, representing friendly or antagonistic relationships
- Positive tie: indicates friendship, trust, or alliance
- Negative tie: indicates enmity, distrust, or conflict
- Balance theory: suggests that triads tend to evolve toward a balanced state, where the product of the signs of the edges is positive
- Balanced triads: all positive edges or two negative edges and one positive edge
- Unbalanced triads: one negative edge and two positive edges or all negative edges